Facilities, Equipment, and Other Resources

This section describes the resources that the University of California, Santa Barbara (the lead organization), Arizona State University (the subrecipient), and the project's collaborating organizations will make available to RedTwin. Only resources that are directly applicable to the proposed work are described. The organizations named in Section 3 receive no funds under this award.

1. University of California, Santa Barbara (lead organization)

UC Santa Barbara is a member of the Association of American Universities and a Carnegie R1 institution. Its College of Engineering and Department of Computer Science provide office and laboratory space, computing, and administrative support to the PIs and their graduate researchers.

Laboratory space. PIs Vigna and Kruegel direct the Security Laboratory (SecLab), located on the second floor of Harold Frank Hall. SecLab provides desk and meeting space for up to twenty researchers, and it includes a machine room with the rack space, power, and cooling required by the server clusters this project will use.

Computing for twin construction and analysis. The proposed work instantiates many virtualized hosts at once, since each digital twin reproduces a deployment of roughly twenty to one hundred components, and it sustains fuzzing and large-language-model inference against those twins on a nightly and weekly cadence. Three tiers of resource cover this. SecLab operates its own Linux servers and workstations for day-to-day development and for the smaller reference networks. The Department of Computer Science maintains a high-performance computing server of four nodes carrying eight NVIDIA A100 GPUs each, which supports model inference and the component-level analyses of Thrust 3. Beyond the department, the Center for Scientific Computing (CSC) at the California NanoSystems Institute operates clusters that are open to any researcher on campus. The Pod cluster provides 40-core CPU nodes for general-purpose computing together with fifteen quad-GPU NVIDIA V100 nodes connected by SXM and NVLINK, and the Knot7 cluster serves large parameter sweeps and jobs that exceed a local workstation. Both are available at no charge to campus researchers, subject to acknowledgment in resulting publications, and CSC staff hold backgrounds in science or mathematics and assist users with porting, developing, and optimizing code. The scale-out runs of Section 7 will use Pod to instantiate the larger twins and to execute the planner against them, while its GPU nodes carry the inference load of the agent teams alongside the departmental A100 server. The UCSB campus cloud, part of the NSF Aristotle project, provides additional elastic capacity.

Isolated environment for digital twins. Section 6 of the Project Description commits to holding every twin on an isolated network, restricting access to the project team and the owning organization, and destroying twins at the end of an engagement. SecLab meets that commitment with an isolated network on reserved IP space separate from the UCSB campus network, and its administrators have long experience running experiments that must stay contained, from malware analysis to the reproduction of vulnerabilities.

Reference infrastructure. The Global Accessible Test Infrastructure (GATE), developed and maintained under the ACTION AI Institute, provides emulated topologies representing smart cities, chemical plants, water systems, and manufacturing plants, and is available to the research community. GATE topologies will serve as input reference infrastructures for the evaluation of Section 7. No funds are requested for them under this award.

Personnel provided through the Early Research Scholars Program. The Early Research Scholars Program (ERSP) is a year-long, dual-mentored research apprenticeship in the Department of Computer Science that pairs small undergraduate teams with a faculty PI and a graduate mentor. Its director, Ziad Matni, has committed to placing a dedicated team of three to five second-year undergraduate researchers with PIs Vigna and Kruegel, co-advised by an assigned SecLab graduate mentor. These students will contribute to curating candidate open-source infrastructure configurations, to designing test cases that exercise the logic-based invariant checks and the attack-graph preservation analysis of Thrust 2, to the empirical evaluation of the vulnerability-chaining agents, and to the design of the twinCTF challenges described in Section 11. This effort is provided at no cost to the award.

Campus systems for the operational evaluation. Sections 6 and 7 describe the evaluation of RedTwin on internal research networks at UCSB, performed only under written authorization that specifies the scope, the permitted observation techniques, and the rules of engagement. This evaluation will be performed in collaboration with the UCSB Information Technology group, and specifically with the Campus Cybersecurity Services group.

2. Arizona State University (subrecipient)

Arizona State University is a member of the Association of American Universities and a Carnegie R1 institution with very high research activity.

Center and laboratory. The Center for Cybersecurity and Trusted Foundations (CTF), part of ASU's Advanced Capabilities for National Security Institute, is a multidisciplinary center whose research spans binary analysis, vulnerability analysis, malware attribution, automated binary analysis systems, and the collaboration of human analysts with automated cyber reasoning systems. Its projects are driven by real-world challenges and result in open-source prototypes. PI Doupé directs the Center and co-directs, with Dr. Shoshitaishvili, the Laboratory of Security Engineering for Future Computing (SEFCOM), which occupies a designated experimental research area within it. Beyond the hardware below, CTF supplies the interdisciplinary connections through which this project can reach collaborators in adjacent areas.

Equipment applicable to this project. SEFCOM operates approximately forty high-end servers and thirty PC systems running Windows and UNIX variants. These provide the ASU-side capacity for the component-level analysis of Thrust 3 and for the mock synthesis of Thrust 2. SEFCOM also holds an ARM DSTREAM high-performance debug and trace unit, which supports the firmware re-hosting that ASU contributes to Thrust 2 for components that cannot be replicated directly.

Educational infrastructure. PI Doupé is a lead developer of pwn.college, and PIs Doupé and Bao are Associate Directors of ASU's American Cybersecurity Education Institute, which operates the publicly available pwn.college instance and facilitates classes for colleges across the United States. This platform will host the open-source infrastructure-security modules described in Section 11, giving them the same distribution as the platform itself.

Campus systems for the operational evaluation. ASU's CTF has excellent relationships with ASU's University Technology Office (UTO). These relationships will enable us to create digital twins of production, enterprise-level applications and deployments. In addition, CTF will work closely with ASU's Chief Information Security Officer so that all discovered vulnerabilities are addressed in the production systems.

3. Resources provided by collaborating organizations

The organizations in this section are not included in the budget and will receive no funds under this award. Each has provided a letter of collaboration, and the resources described below are those the letters commit to, on the conditions those letters state.

Sage Bionetworks. Sage Bionetworks develops and maintains Synapse, an open-source repository and collaboration environment for biomedical research data released under the Apache License 2.0. Sage holds data resources of its own, acts as a steward and broker for resources contributed by other institutions and consortia, and maintains the governance machinery, including data-use agreements, consent frameworks, access review, and de-identification practice, that determines what may be shared and under what conditions. For the synthetic-data work of Thrust 2, Sage has indicated that where a specific use case is well defined and has been validated against the applicable consent terms, data-use agreements, and access-review process, it expects to work with the team on access to material suitable for deriving realistic, non-identifying populations for hospital digital twins, meaning structure, distributions, and relationships rather than records. Where direct access is not appropriate, Sage expects to help specify what a defensible synthetic substitute would have to preserve. Any such arrangement is subject to Sage's governance review and to the conditions set by the original data contributors.

The ZMap project. Zakir Durumerić, an original author and continuing maintainer of ZMap, has indicated that the extensions developed under this project, including probe modules, structured output formats, incremental-scan support, and constraint handling, would be welcome through the project's normal review process, so that capabilities built for RedTwin become available to all ZMap users instead of living in a fork. He has also offered advice on scanning methodology, covering rate control, vantage-point selection, probe-module design, and the safeguards that should govern the observation of operational networks. This gives Thrust 1 a maintained upstream path and review capacity the project does not otherwise have.

The NGINX community and F5, Inc. Buu Lam, Director of Community Evangelism for NGINX at F5, operates the channels through which NGINX practitioners exchange operational knowledge. He has offered to help the team reach practitioners willing to describe representative deployment topologies, which is the source of the deployment reality that Thrust 1 requires; to review configuration-level findings and assess whether a given result reflects a pattern common in the field or an artifact of a synthetic test network; and to make NGINX community channels, events, and technical-content programs available for carrying results and artifacts to operators. These arrangements are subject to internal review at F5.

The RabbitMQ project and Broadcom Inc. Gabriele Santomaggio, a member of the RabbitMQ team at Broadcom, has offered maintainer input on the properties that determine whether a replicated broker is faithful, including which aspects of cluster formation and inter-node communication are security-relevant, how virtual-host and permission structures should be represented, which plugin combinations change the exposed surface, and where a superficially correct replica would diverge from a real deployment under load or partition. This supports the clustered-service stress test of Thrust 2 directly. He has further indicated that findings in the code would be handled through the project's normal security reporting process, and that findings about deployment and configuration would inform safer defaults, configuration validation and diagnostics, and corrections to the documentation and reference deployments that users copy. These arrangements are subject to internal review at Broadcom.

4. Other resources

Competition infrastructure. The PIs have organized the iCTF competition annually since 2003. The twinCTF competition of Section 11 will reuse the existing iCTF hosting and scoring infrastructure rather than standing up new infrastructure of its own.

Shared development infrastructure. The two institutions already share a single source repository and continuous integration system for existing joint efforts, and this project will use the same arrangement, as described in Section 6.